A Little Red Dot Lights Up in Radio

Archival radio data reveal a compact counterpart to a local little-red-dot, probing its nature.

arXiv:2610.17559 — radio counterpart of a local little-red-dot

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Background

Little red dots are compact, red, high-redshift galaxies whose nature—accreting black hole or dense star formation—remains debated. Finding a nearby example lets us study one in far greater detail.

Little-red-dot radio emission is rarely detected, so we lack direct probes of jets or AGN activity in these objects.

What this paper is selling

The authors stack ten archival radio epochs at the Gaia position of J092359+415736, detecting a compact radio source. Its SED and radio properties are compared with other reddened AGN and little-red-dot samples.

Key results to remember

- A compact radio source is detected at the Gaia position across multiple archival epochs.
- The SED fits a two-component blackbody with effective temperatures of 4583 K and 2086 K.
- LoTSS shows a 42-sigma compact 144 MHz source, confirming genuine radio emission.

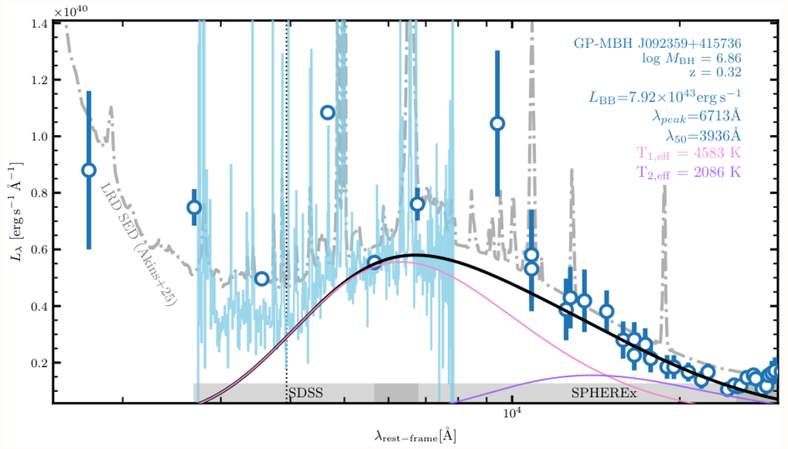


Figure 2 | The two-temperature blackbody fit shows J092359+415736 sits among reddened AGN and little-red-dot samples in luminosity-wavelength space.

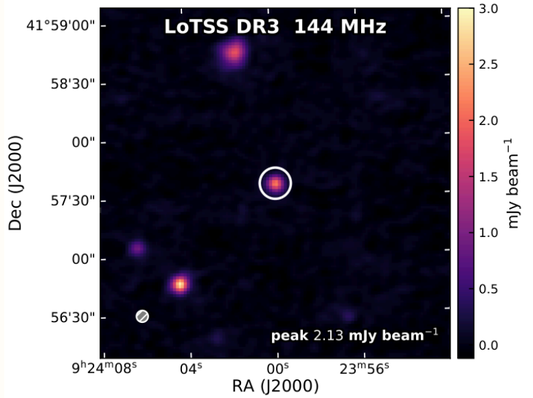


Figure 4 | A 42-sigma compact 144 MHz detection at the Gaia position confirms the little-red-dot has a genuine radio counterpart.